Euclid, Book I, Proposition I.10

To Bisect a Given Finite Straight Line

Formal Reconstruction — Technical Documentation

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1 Introduction

Proposition I.10 constructs the midpoint of a given finite straight line.

In Euclid's development this construction depends on several earlier results. An equilateral triangle is first erected on the given segment using Proposition I.1. The vertex angle of this triangle is then bisected by Proposition I.9, and the Side–Angle–Side congruence criterion of Proposition I.4 is used to prove that the intersection point is the midpoint of the original segment.

The Hilbert reconstruction follows a fundamentally different approach. The existence of a midpoint has already been established as an independent theorem of the Hilbert interface. Consequently, Proposition I.10 becomes a direct application of this previously derived result.

This proposition therefore provides one of the clearest examples of the difference between Euclid's historical development and a modern axiomatic reconstruction.

2 The Classical Configuration

Let AB be the given finite straight line; see Figure 1.

The objective is to construct a point lying on the segment AB that divides it into two congruent parts.

Such a point is called the midpoint of the segment.

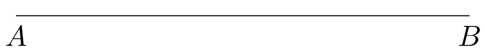

Figure 1: The given finite straight line AB.

2.1 Construction

Construct the equilateral triangle ABC on the segment AB by Proposition I.1.

Next, bisect the angle $\angle ACB$ by Proposition I.9.

Let the angle bisector meet the segment AB at the point D; the completed construction is Figure 2.

The claim is that D is the midpoint of the segment AB.

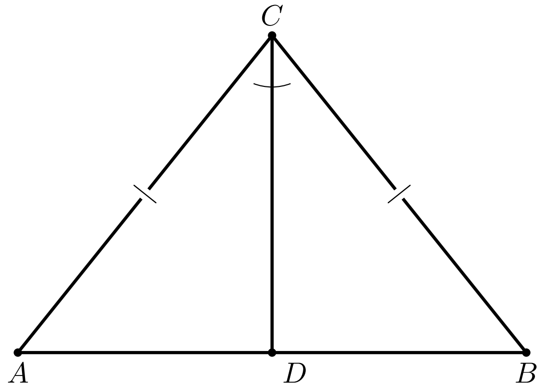

Figure 2: Euclid's construction for Proposition I.10. The equilateral triangle gives $AC \cong BC$; the ray CD bisects the angle at C. SAS for ACD and BCD yields $AD \cong DB$.

2.2 Application of Proposition I.4

Consider the triangles

$$\triangle ACD \quad \text{and} \quad \triangle BCD.$$

Since the triangle ABC is equilateral,

$$AC \cong BC.$$

The segment

$$CD$$

is common to both triangles, and by construction,

$$\angle ACD \cong \angle DCB.$$

Therefore the hypotheses of Proposition I.4 are satisfied.

It follows that

$$AD \cong DB.$$

Since the point D lies on the segment AB, it is the midpoint of the given finite straight line.

3 Statement

Theorem 1 (Euclid I.10)**.** *Let AB be a finite straight line with*

$$A \neq B.$$

Then there exists a midpoint of the segment AB. That is, there exists a point M such that

$$A, M, B$$

are collinear,

$$M$$

lies between A and B, and

$$AM \cong MB.$$

Equivalently, every nondegenerate segment admits a midpoint.

Diagrammatic audit. Two geometric figures are sufficient. The first records the input segment; the second contains the entire Euclidean construction and the data used by SAS. Wilkins likewise presents the equilateral-triangle/angle-bisector configuration as the essential picture. His note also records a distinct historical construction attributed by Proclus to Apollonius, using two circles and SSS; that alternative is mathematically interesting but is not part of Euclid's proof adopted here and is therefore not duplicated as a second construction.

4 Classical Proof

Construct the equilateral triangle ABC on the segment AB by Proposition I.1.

Bisect the angle $\angle ACB$ by Proposition I.9, and let the bisecting ray meet the segment AB at the point D.

Since

$$AC \cong BC,$$

$$CD \cong CD,$$

and

$$\angle ACD \cong \angle DCB,$$

the triangles

$$\triangle ACD \quad \text{and} \quad \triangle BCD$$

satisfy the hypotheses of Proposition I.4.

Hence

$$AD \cong DB.$$

Because the point D lies on the segment AB, it is the midpoint of the given finite straight line.

This completes the construction.

5 Proof Architecture of the Reconstruction

The Hilbert reconstruction replaces the entire classical construction by a previously established theorem of the Hilbert interface.

Rather than constructing an auxiliary equilateral triangle, bisecting an angle, and applying the Side–Angle–Side criterion, the existence of a midpoint is obtained directly from the general midpoint existence theorem.

Consequently, Proposition I.10 becomes a simple wrapper around an existing interface theorem.

```
Euclid
------
given segment AB
      |
I.1: equilateral triangle ABC
      |
I.9: bisect angle ACB
      |
I.4: SAS
      |
AD ~= DB
      |
D is midpoint of AB


Hilbert / Lean
--------------
A != B
  |
  v
HilbertMidpointExists
  |
  v
exists M, HilbertIsMidpoint Geo M A B
  |
  v
euclid_proposition_10
```

6 Hilbert Reconstruction

The reconstruction consists of a direct application of the general midpoint existence theorem.

Given a nondegenerate segment, the theorem `HilbertMidpointExists` immediately produces a point lying between the endpoints and equidistant from them.

No auxiliary geometric construction is required, since the existence of midpoints has already been established independently within the Hilbert development.

This represents one of the largest simplifications obtained by the layered architecture of the Hilbert interface.

7 Lean Formalization

The Lean theorem is correspondingly short:

```
theorem euclid_proposition_10
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (A B : Geo.Point)
    (hAB : A != B) :
    exists M : Geo.Point,
      HilbertIsMidpoint Geo M A B := by
  exact
    HilbertMidpointExists
      Geo A B hAB
```

Thus the proposition itself introduces no auxiliary construction. Its mathematical content has already been packaged in `HilbertMidpointExists`; I.10 is the Euclidean-facing wrapper exposing that result in the Book I development.

8 Relationship with Earlier Propositions

8.1 Proposition I.1

Euclid's construction of the midpoint begins by constructing an equilateral triangle on the given segment. Proposition I.1 therefore supplies the first auxiliary point and the equal sides needed later in the congruence argument.

8.2 Proposition I.4

The decisive classical step is SAS. After the angle at the apex has been bisected, the two resulting triangles share one side, have a pair of equal sides from the equilateral construction, and have congruent included angles. Proposition I.4 then yields congruence of the two triangles and hence equality of the two halves of the original segment.

8.3 Proposition I.9

Proposition I.9 is the immediate constructional input: it bisects the angle at the apex of the equilateral triangle.

Thus Euclid's dependency chain is particularly transparent:

$$\text{I.1} \longrightarrow \text{I.9} \longrightarrow \text{I.4} \longrightarrow \text{I.10}.$$

8.4 Transition to Proposition I.11

The midpoint construction becomes useful immediately in the classical proof of Proposition I.11, where a symmetric configuration about a point on a line is used to construct a perpendicular.

More generally, midpoint and segment-bisection infrastructure becomes a recurring component of later Book I constructions.

9 Hilbert and Book Zero Components Used

9.1 Midpoint Representation

The formal conclusion is expressed by the project midpoint predicate: the constructed point lies between the endpoints and the two resulting segments are congruent. Thus midpoint remains entirely synthetic:

$$A - M - B \qquad \text{and} \qquad AM \cong MB.$$

9.2 Interface Theorem

At the level of Proposition I.10 itself, the only substantive geometric dependency is

```
HilbertMidpointExists
```

together with the incidence and congruence structures required by that theorem. Angle bisection and SAS belong to Euclid's historical proof; they are not invoked by `euclid_proposition_10`.

10 Formalization Notes

Proposition I.10 provides one of the clearest illustrations of the difference between Euclid's historical development and a layered axiomatic reconstruction.

In the *Elements*, the midpoint construction depends on three earlier propositions. Euclid first constructs an equilateral triangle, then bisects one of its angles, and finally applies the Side–Angle–Side criterion to prove that the resulting point divides the original segment into two equal parts.

The Hilbert reconstruction is fundamentally different. The existence of midpoints has already been established as an independent theorem of the Hilbert interface. Consequently, Proposition I.10 becomes a direct application of this theorem, requiring no additional geometric construction.

This proposition demonstrates the principal advantage of the layered architecture developed in CGJteamLab. Once sufficiently rich interface theorems have been established, many classical propositions cease to be independent proofs and instead become concise applications of the existing theory.

Proposition I.10 is perhaps the simplest example of this philosophy, reducing an entire classical construction to a single theorem call.

11 Dependency Summary

The two proof architectures should be kept distinct:

```
Classical:
I.1  ->  I.9  ->  I.4  ->  I.10

Formal:
HilbertMidpointExists  ->  euclid_proposition_10
```

The formal theorem does not replay Euclid's equilateral-triangle, angle-bisection, and SAS construction. It reuses a midpoint theorem already proved in the Hilbert interface. No coordinate system, numerical length, or new geometric axiom is introduced.

12 Conclusion

Proposition I.10 completes the first ten propositions of Book I with the construction of a midpoint.

The classical proof is a compact synthesis of three earlier results: I.1 supplies an equilateral triangle, I.9 bisects its apex angle, and I.4 converts the resulting SAS configuration into equality of the two halves of the given segment.

The formal reconstruction preserves this synthetic character while making one feature especially explicit: a midpoint is not merely a point with two equal distances. Its position between the endpoints is part of the theorem and must be carried by the Hilbert order structure.

I.10 therefore consolidates the early construction and congruence machinery into a reusable segment-bisection result. This midpoint infrastructure becomes immediately relevant to the perpendicular constructions beginning with Proposition I.11.